Powder Technology 228 (2012) 416–423

Contents lists available at SciVerse ScienceDirect

Powder Technology

journal homepage: www.elsevier.com/locate/powtec

Review

A review of the Residence Time Distribution (RTD) applications in solid unit operations

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ARTICLE INFO

Article history:
Received 30 January 2012
Received in revised form 21 May 2012
Accepted 26 May 2012
Available online 9 June 2012

Keywords:
Solids
RTD
Unit operation
Modeling
Measurement
Performance

ABSTRACT

This review traces current applications of the residence time theory in various solid unit operations. Besides reviewing recent experimental and simulation studies in the literature, some common modeling and tracer detection techniques applied in continuous flow systems are also considered. We attempt to clarify and emphasize the influence of the residence time profile on the unit performance, which is the key in system design and performance improvement of practical unit operations. The development of predictive modeling is also an important goal in the long-term development of the residence time theory.

Contents

1. Introduction

In chemical engineering and related fields, the Residence Time Distribution (RTD) is defined as the probability distribution of time that solid or fluid materials stay inside one or more unit operations in a continuous flow system. It is a crucial index in understanding the material flow profile, and is widely used in many industrial processes, such as the continuous manufacturing of chemicals, plastics, polymers, food, catalysts, and pharmaceutical products. In order to achieve satisfactory output from a specific unit operation, raw materials are designed to stay inside the unit under specific operating conditions for a specified period of time. This residence time information is usually compared with the time necessary to complete the reaction

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doi:10.1016/j.powtec.2012.05.060

or process within the same unit operation. For example, in continuous powder mixing processes, powder is mixed in a continuous mixer. The local mixing rate coupled with the time the powder stays inside the mixer determines the unit performance. If the time required for local mixing is longer than the actual residence time powder stays in the system, the process cannot provide a complete mixture, and it fails its designed purpose [1]. In other words, the performance of any continuous unit operation is determined by the competition of the two sub-processes: a batch process or reaction superimposed by an axial flow. Therefore, the characterization of the RTD in different continuous unit operations is the first step in the design, improvement, and scale-up of many manufacturing processes in the chemical engineering industry.

The research on the RTD in chemical engineering fields has focused on the influence of operation conditions, materials, and the unit geometry on the RTD profile, the improvement of measurement methods, and the improvement of predictive modeling on different processes and units. Most studies investigated continuous unit operations by using the RTD; few extended to the correlation between the RTD and the reaction or process performance, which is usually case-sensitive. For example, a continuous polymer foaming process was studied in an extruder [2], where the thermal decomposition rate of chemical blowing agent was compared with the RTD to investigate the optimization of the foam density; a chemical-looping combustor was investigated in both continuous and batch mode, in which the RTD was used to develop a model for predicting the mass-based reaction rate constant for char conversion [3]; the production of polypropylene was characterized in a horizontal stirred bed reaction by considering the RTDs of catalyst and polymer separately, which strongly depend on the temporal catalyst activity [4]; the emulsification process in polymer mixing was studied in a twin-screw continuous extruder, where the RTD and the morphology profile of the mixture was examined simultaneously in one pulse test [5]; the Cr(VI) reduction in wastewater treatment was investigated in an electrochemical tubular reactor by applying CFD and velocity profile, where the performance was coupled with the axial flow rate [6].

Due to the wide scope of the RTD issue, every year a large number of papers have been published using this conception in a host of disciplines. A previous review by Nauman [7] summarized the theoretical development history of the RTD since the beginning of the last century, especially for continuous fluid systems. Some developments have occurred since the previous review that will be covered in this paper. Moreover, this review mainly profiles the applications of the RTD theory in characterizing solid chemical engineering unit operations. Also, this review differs from the previous one in that it emphasizes coupling the RTD with the unit performance of specific unit operations. This paper is organized in the following manner. Section 2 illustrates different modeling work of the RTD in these applications, followed by a discussion on general RTD measurement methods in Section 3. In Section 4, recent applications of the RTD in solid continuous flow and manufacturing systems are described in details. Our goal in this review paper is to bring together the recent applications of the RTD theory across a wide range of studies in the chemical engineering fields, and contribute to the performance investigation of versatile continuous unit operations.

2. RTD modeling

2.1. CSTR and PFR series

Modern RTD theory originally developed from continuous fluid systems [8]. Early fluid reactor models assumed plug flow in a tubular-shape reactor (PFR), or perfect mixing in continuous stirred tank reactors (CSTR). These conceptions represent two extreme RTD profiles in the reactor. In practical continuous flow systems, experimental RTD profiles are usually between the two extremes. To describe the non-idealness of the RTD profile, different combinations of CSTR and PFR were introduced in modeling practical cases. CSTR in a series model is one commonly used model [3,9–11]:

$$\tau = NV_0/F \tag{1}$$

$$E(\theta) = \frac{N(N\theta)^{N-1}}{(N-1)!}\exp(-N\theta) \tag{2}$$

where τ is the mean residence time (MRT), N the number of CSTR tanks, V_0 the volume of each tank, and F the volumetric flow rate. $E(\theta)$ represents the dimensionless RTD and $\theta = t/\tau$ the dimensionless time. As a one-parameter model, the idealness of the RTD is represented by the number of CSTR tanks used (Fig. 1). Large number of tanks indicates a PFR-like reactor ($N \to \infty$), and a small number leads to a CSTR-like reactor ($N = 1$). Two modifications were reported: the tanks in series were followed by a PFR element in case of the RTD rise part delay when axial dispersion is significantly limited [2,12]; backward flux was introduced among the CSTR tanks to capture the long tail in the RTD profiles [4,13]. In the second modification, large backward flux indicates fast material exchange between adjacent tanks. The long tail profile can also be modeled by a dead zone volume cross-flowing with the CSTR element [14–16]. Notice that the fraction of the dead zone element represents the degree of non-idealness of the continuous flow system. Amador et al. [17] reported a resistance network model that can be considered as parallel connections of a series of PFR elements [18], thus also belonging to this kind of model.

2.2. Axial dispersion model

Despite the combination of CSTR and PFR elements, the axial dispersion model is an efficient alternative to generalize the conception of the RTD to most non-ideal reactors. The differential equation representing the axial dispersion process of a tracer in the flow system, or the Fokker–Planck equation (FPE), is expressed as a global 1D equation:

$$\frac{\partial c}{\partial t} = E\frac{\partial^2 c}{\partial z^2} - v_z\frac{\partial c}{\partial z} \tag{3}$$

where c is the concentration of a component in the system, v_z is the global axial velocity, and E denotes the dispersion coefficient in solid systems, or the diffusion coefficient in fluid systems, respectively. The variables t and z represent time and axial distance from the tracer injection point. The advantage of the FPE is the clear physical meaning of its parameters, where v_z and E indicates the combination of the axial transport and the superimposed axial dispersion in this model. There have been many industrial studies on the RTD directly using the FPE [13,19–23]. Based on previous literature [24,25], Sherritt et al. [26] summarized various FPE solutions under different boundary conditions in a rotary drum. For example, Vashisth and Nigam [27] applied the solution based on error function to analyze single-phase laminar flow through a straight tube. The most widely used solution of the RTD was developed by Taylor [28] with open–open boundaries:

$$E(\theta) = \frac{1}{2\sqrt{\pi\theta/Pe}}\exp\left\{-\frac{Pe(1-\theta)^2}{4\theta}\right\} \tag{4}$$

where $\theta = t/\tau$ is the dimensionless time, τ is the mean residence time (MRT); $Pe = v_z l/E$ represents the Peclet number, and l is the distance between injection and detection points. This solution was applied in the fitting of experimental RTD data in various systems [1,9,29–31]. Although some of the reported systems were not single phase or

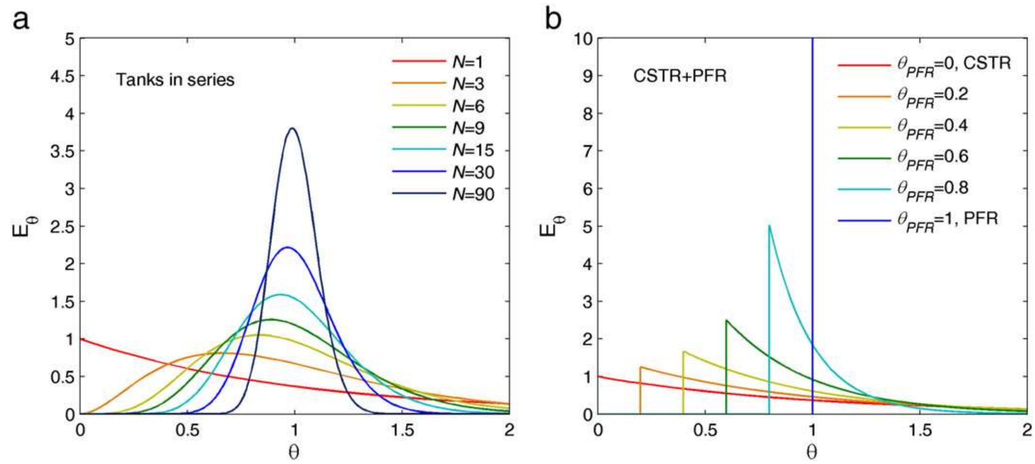

Fig. 1. E curves of some RTD models reviewed in this work.

with open–open boundary conditions as when this solution was derived, good agreement was obtained, especially for rise-delayed or long-tailed RTD profiles, showing the robustness of the Taylor's dispersion model. A more convenient way to estimate the Peclet number without RTD curve fitting is using the following formulas on the experimental RTD data:

$$\tau = \int_0^\infty tE(t)dt \tag{5}$$

$$\frac{\tau}{\sigma} = \sqrt{\frac{2}{Pe} - \frac{8}{Pe^2}} \tag{6}$$

where $E(t)$ represents the RTD as a function of time, and σ is the second moment of the RTD:

$$\sigma^2 = \int_0^\infty (t-\tau)^2 E(t)dt \tag{7}$$

Recent applications of Eq. (6) can be found in the work of Sudah et al., Vikhansky, Fang et al., Kumar et al., and Waje et al. [23,32–35]. In granular systems, a unique method to estimate the dispersion coefficient was reported by using information of single particle movement [22,36,37]:

$$E = \lim_{\Delta t \to 0} \frac{\langle \Delta x^2 \rangle - \langle \Delta x \rangle 2}{2\Delta t} \tag{8}$$

where Δx^2 is the square displacement of a single particle within the time interval Δt. The symbol $< >$ indicates the statistical expectation of the parameter within. As these studies indicate, the prediction of the axial dispersion coefficient is the key step for the development of predictive RTD models.

2.3. Stochastic model and Markov chain

Another widely used RTD model is to consider the particle movement as a stochastic process. In particular the axial motion of any particle or small fraction of material inside a continuous system is considered as a two dimensional probabilistic process. As a result, the overall RTD curve can be calculated by the accumulation of the random axial motions of these small fractions. It can be applied to a process where the predictive RTD model is absent, for example, due to complicate system geometry [38]. One stochastic model that is similar to the CSTR and PFR series is the Markov chain model. This model arbitrarily defines the ratio of flux exchange among assumed elements connected in a network, and the next state of the system depends only on the current state and nothing else:

$$S(n+1) = PS(n) \tag{9}$$

$S(n)$ is a $1 \times m$ state vector of the system, describing the material distribution at time t_n, where m is the total number of elements in the system; P is a $m \times m$ matrix of transition probabilities for the time interval between two adjacent states. Although the exchange of flux is well modeled among different elements, mixing inside each element is not considered in a Markov chain model, which is different from the CSTR and PFR series model. A detailed description of the Markov chain can be found in [39]. This model is very useful when the flow regions within the system can be easily distinguished into different elements, for instance, the granular flow with heterogeneous cross-wise layers [16,40], the gas flow at different regions in an entrained flow gasifier [41], and in fluidized beds [42,43].

2.4. Bimodal RTD

Bimodal RTD profile with dual peaks is not common in the literature. It results from two or more main flow components traveling differently, for instance, through two paths in reactor [9,44], or on different layers in a granular flow system [40]. The dual peaks result from the overlap of RTD components through different paths [27]. Due to case-sensitivity, no quantitative model has been developed for bimodal RTD profile.

2.5. Convolution

In fluid systems, the non-idealness of a tracer injection is common due to a relatively short MRT compared to the injection period of a fluid pulse. To correct this non-ideal RTD, recent studies applied the algorithm of convolution, introduced by Danckwerts [8], as the injection was usually not a perfect pulse [9,30,45,46]:

$$E_{out}(t) = \int_0^t E_{in}(t-\theta)E(\theta)d\theta \tag{10}$$

or

$$E_{out}(t) = E_{in}(t) \otimes E(t) \tag{11}$$

where $E_{out}(t)$ and $E_{in}(t)$ represent the RTD curve before correction and the imperfect pulse injection detected at the input of the system.

The convolution integration corresponds to multiplication in the frequency domain, therefore:

$$FFT\{E_{out}(t)\} = FFT\{E_{in}(t)\}FFT\{E(t)\} \tag{12}$$

$$E(t) = FFT^{-1}\left\{\frac{FFT\{E_{out}(t)\}}{FFT\{E_{in}(t)\}}\right\} \tag{13}$$

where FFT and FFT^{-1} represent the fast Fourier transform and the inverse Fourier transform operations. In continuous blending systems, this knowledge was also applied in characterizing the attenuation of feeding fluctuations [29]. Another application of convolution was reported in the assemble of the overall RTD profiles from the RTDs of single elements in an extruder study [47]:

$$E(t) = E_1(t) \otimes E_2(t) \otimes \cdots \otimes E_n(t) \tag{14}$$

Similar RTD assemble studies can be found in Gao et al. and Essadki et al. [1,9].

2.6. RTD constructed by velocity profile

Various CFD software and velocity sensors were used to give velocity profiles in fluid systems, which were further applied to construct RTD profiles [6,27,44,48,49]. In solid systems, the equivalent tools are the Discrete Element Method (DEM) and positron emission particle tracking (PEPT) [22,36,50]. However, due to the variability of the velocity profile in most solid unit operations, the RTD evaluation based on practical velocity profile is not common. In one specific case, the convection model was specifically used in describing the RTD profiles of laminar tubular flow. The axial dispersion is constructed from the well-known parabolic laminar velocity profile in the tube by applying classic fluid mechanics. The MRT is the spatial time $\tau = Q/V$ where Q is the volumetric flow rate and V is the internal volume of the tube. The application of a generalized form of the convection model can be found in both fluid and solid extrusion systems [45–47].

3. RTD measurement

The stimulus response test is a common method used in most experimental RTD measurement. In this test, pulse injection or step change of a tracer is performed at the inlet of a continuous system where steady state of bulk flow is reached, and response of the tracer profile at the outlet is recorded. The selected tracer is expected to share similar properties with the bulk material, thus introducing as few disturbances as possible on the bulk flow. For example, Harris et al. [51] applied a phosphorescent pigment as the bulk flow in a circulating fluidized bed (CFB), and the tracer was generated by activating the same kind of pigments. Thus, the tracer material is identical to the bulk material, and hydrodynamic flow disturbances were avoided. Another assumption of stimulus response test is a closed inlet and outlet boundaries. This guarantees unidirectional flow and no boundary dispersion. Exceptionally, in some systems with open–open boundaries, boundary dispersion can also be considered negligible when axial transport is much faster than the dispersion (for example, with large Peclet number). Therefore, the stimulus response test developed under the assumption of closed boundaries can be applied in some open systems.

To our knowledge two types of tracer detection techniques have been reported: inline and offline detection. Inline detections directly record the optical, thermal, or electrical signal of tracer concentration from inline probes, and retain the data automatically for further analysis. It requires fast sample acquisition, signal conversion, and data storage processes. NIR spectroscopy was reported for inline monitoring of drug concentration in a continuous powder blender [52]. When one or more of these steps can not be finished immediately, offline detection is selected. For instance, digital imaging analysis can hardly be inline due to the slow extraction of tracer concentration from digital photos [4,23,34,39,47,53]. Most tracer detections in solid processes were also offline as the act of solid sampling is usually invasive, and can not be very fast [54]. The treatment of solid samples, like dissolution of dye or salt tracers, also takes time, leading to offline detections [14,15,20,21,35].

The tracer detection can be performed optically or conductively. Mainly two kinds of optical detection systems were reported. Digital image processing and color meter were used to analyze the color of photos or the signal from the sensor directly [11]. Fluorescence analysis also belonged to this kind of detection, which was applied more in dark environment with accurate sensors [5,30,33]. Another optical detection reported in many literatures utilizes different types of spectroscopy to analyze the light absorbance of the tracer, for example, Parker blue dye dissolved in water [55]. Near infrared (NIR) and ultraviolet (UV) are two commonly used regions of the spectrum, reported by many investigators [12,14,20,27,35,54,56,57]. On the other hand, conductive detection systems are based on the difference of electrical conductivity between tracer and bulk flow, for example, NaCl in deionized water [9,45,46,49,58]. Sensitive detector based on the difference of gas thermal conductivity was also reported in a gas fluid system [19]. Helium was used as the inert tracer in this study. Hot or radioactive particles have been applied as tracer injection and detected using thermistors or scintillation detectors in gas–solid fluidization systems [59–61]. Tracer using organic gas such as propane was also reported, the detection of which was using inline gas chromatography in a CFB system [13].

Besides the stimulus pulse test, particle tracking was reported as an alternative RTD measurement method. Technique such as PEPT, and the application of DEM or CFD simulations provide movement and residence time information of a huge number of particles, and the RTD can be derived from these particle tracking information. Since no tracer is involved, this method does not introduce the problem of tracer disturbance. However, an additional difficulty comes from the amount of information that has to be recorded from intensive particle tracking experiments or computations, as well as the accuracy of experiments and simulations. The computational data should be validated with experiments before the simulated RTD profile can be applied in practice. Details of efforts on this issue can be found in [17,22,36,48,50].

4. RTD applications in solid process

4.1. Continuous blender

As much of the initial research on continuous powder blending came from the extension of continuous liquid blending based on the response-stimulus test, the attenuation effect of the RTD on feeding fluctuations was well studied in early theoretical development [8]. The axial mixing component of continuous powder mixing was clarified by applying Eqs. (9–12) [29,62]. These studies on continuous powder blending were summarized in the review by Pernenkil and Cooney [63], which also placed emphasis on the similarity of blending mechanisms between batch and continuous systems. This similarity, described as the cross-sectional mixing component of continuous blending, was verified by using the RTD or the mean residence time (MRT) to link the continuous blending performance with the corresponding batch system under similar operation and geometry [1,64]:

$$\sigma_c^2(z) = \int_0^\infty \sigma_b^2(t)E(t,z)dt \tag{15}$$

where $\sigma_c^2(z)$ indicates the variance decay in the continuous blender as a function of location z; $E(t,z)$ is the RTD measured at location z inside the continuous blender; $\sigma_b^2(t)$ is the variance decay in an equivalent batch blender as a function of time. Due to the exponential decay relationship of variance in most batch blenders:

$$\sigma_b^2(t) = \sigma_{ss}^2 + \left(\sigma_0^2 - \sigma_{ss}^2\right)\exp(-k_b t) \tag{16}$$

Eq. (15) can be simplified as the following formula:

$$\sigma_c^2(z) = \sigma_{ss}^2 + \left(\sigma_0^2 - \sigma_{ss}^2\right)\exp(-k_c z) \tag{17}$$

where σ_0^2 and σ_{ss}^2 are the initial and steady state variance of mixture; k_c the variance decay rate in the continuous blender, and k_b the decay rate in the equivalent batch blender. The relationship between k_c and k_b is

$$k_c = k_b / v_z \tag{18}$$

Eq. (18) indicates the fundamental principle of continuous blending: fast batch-like mixing rate and slow axial motion (or long residence time) lead to good mixing performance.

Due to the importance of the RTD in the characterization of continuous solid blending, recent work applied it on both mixing components described above. Sherritt et al. [26] illustrated a method for axial dispersion prediction in rotary drum reactors for both batch and continuous mixers. Marikh et al. [39] examined the correlation between operations and hold-up, MRT, and axial dispersion in a convective mixer. The cross-sectional mixing efficiency was studied using the fluctuation of the RTD measurement, and a correlation between mixing performance and the fitting error of the RTD measurement was established [57]. In a previous paper from our group [54], the effects of feed rate, blade angle, and rotary speed was investigated in a convective continuous blender. The RTD was used to correlate with the mixing performance under different conditions. One interesting point in this work is the introduction of the number of blade passes, the product of the rotary speed and the MRT. This term describes the total number of revolutions that powder encounters in the continuous system [65]. Larger blade passes usually indicates better mixing performance at constant hold-up. Portillo et al. [22] also studied particle motion in a continuous mixer using PEPT. Based on the information of tracer trajectory, the MRT and axial dispersion coefficient was calculated, and a linear correlation was found between trajectory length and the residence time of a single particle. The RTD of cohesive particles was examined in one periodic slice of a convective mixer in a discrete element method (DEM) environment [50]. The horizontal stirred bed reactor for polymer production was investigated [3]. While the polypropylene production was provided by blending between polymer and catalyst, polymer was fed continuously along the reactor. This is equivalent to a batch blending process in which the polymer was added continuously.

4.2. Extruder

Recent extrusion studies mainly investigated the RTD in polymer and food manufacturing processes. Much work focused on the operation of co-rotating twin-screw extruders or single screw extruders. Polymer flow behavior was studied by Fang et al. [33] in a co-rotating twin-screw extruder, where polymers with higher viscosity or under high pressure showed longer residence time. Baron et al. [20] developed a model for RTD prediction in fully intermeshing extruders based on the extension of an axial dispersion model, including operations (screw speed and flow rate) and geometries (screw profile and die design). In a twin-screw polymer mixing process, Zhang et al. [5] applied the conception of the RTD into a simultaneous detection of polymer morphology of mixture with different component ratios. Ziegler and Aguilar [11] studied the continuous processing of chocolate in a twin-screw extruder, and the mean residence time was found inversely proportional to screw speed while directly proportional to feed rate. The single screw study on rice flour by Yeh and Jaw [15] and the studies by Bi et al. [53] and Kumar et al. [34] led to the same results. In the single screw dryer studied by Waje et al. [35], which conserved the structure of screw but was not an extruder, similar effects of feed rate and screw speed were observed.

Polymer foam process was investigated based on the separate consideration of the batch-like chemical blowing agent foam process and the RTD in the extrusion process [2]. A gear pump associated with the single screw extruder was used to adjust the RTD, and thus the density of the foam in the polymer form process. In another study of polymer form process, the injection of supercritical carbon dioxide was used as physical forming agent [14]. Results showed that high screw speed or high temperature implies short residence time while the form performance was not discussed in that study.

One interesting RTD study using a finite element method in a reciprocating single-screw pin-barrel extruder was reported by Bi and Jiang [47], where the RTD of the overall extruder was assembled from the RTD simulations of single screw elements by the convolution technique. This work provides flexibility based on the existing geometry design method of continuous extrusion process.

4.3. Rotary drum

Rotary drum was widely used in blending, calcination, drying, granulation, and many other solid unit operations. Current RTD studies on rotary drums have roots in the work by Saeman [66], where the predictive model of mean residence time, bed depth distribution, and other characteristics related to the axial motion of a solid was given for a rotary drum without flights. According to this model, the bed depth h at the axial position z can be described by the following differential equation:

$$\frac{dh}{dz} = \frac{3\tan\gamma}{2\omega} F\left[R^2 - (h-R)^2\right]^{-3/2} + \frac{dR}{dz} - \frac{\tan\beta}{\cos\gamma} \tag{19}$$

where γ is the repose angle of the solid material, F the volumetric flow rate; ω is the rotary speed (for example, rad/s), β the incline angle, and R the inner radius of the drum. The bed depth at the outlet of the drum can be used as boundary condition of the differential equation, which equals to the dam height or the average diameter of the particles:

$$h(z=0) = h_{dam}\, or\, h(z=0) = d_{particle} \tag{20}$$

By solving Eq. (19), the volumetric hold up in the drum can be integrated from the bed depth profile

$$V_{holdup} = \int_0^L R(z)^2[\theta(z) - \sin\theta(z)\cos\theta(z)]dz \tag{21}$$

$$\theta(z) = \arccos\left[1 - \frac{h(z)}{R(z)}\right] \tag{22}$$

Thus the MRT can be calculated as

$$\tau = V_{holdup}/F \tag{23}$$

Notice that Saeman's model cannot provide prediction for the axial dispersion inside the drum. McTait et al. [16] developed a model to describe the particle motion, correlating the cross-sectional rolling of particles with their axial motion. A theoretical estimation of the axial dispersion from the number of particle rotation was derived and validated. By using sand, a recent study by Liu and Specht [67] verified Saeman's model under various feed rates and rotation speeds, and later Liu et al. [68] developed an analytical solution for Saeman's model for cases with fill level lower than 25%. Experimental results indicated that the MRT is inversely proportional to the rotation speed, and slightly increases with the feed rate. These results were the same as the work by Sudah et al. [23], where experiments were conducted to determine the effects of operations on the overall hold-up and the RTD profiles of cylindrical zeolite catalysts. While the experimental values of hold-up and the MRT were close to the predictions from the Saeman's model, the axial dispersion results were proportional to rotary speed and incline angle and was limited when feed rate and fill level were large enough. A recent study by Chen et al. [69] drew similar conclusions, which emphasized more on the MRT and the mean flow rate (MFR) under various conditions. The MFR symbolized the productivity and the economic efficiency of a rotary drum.

Scott et al. [70] carried out experiments in an inclined rotary drum fitted with various dams, focusing on the effects of the dam on the bed depth distribution and particle flow pattern. Saeman's model was still applicable in this case, which provided good predictions for bed segments with different boundary conditions. The results are useful for industrial kilns where conical brickwork dams at different locations along the kiln are common. For rotary drums with flight, the Saeman's model was not applicable anymore. The flow of particles in this kind of geometry is similar to the convective mixer discussed above. Cronin et al. [38] introduced stochastic modeling to account for a larger axial dispersion due to the random effects of flight sweep. Particle motions between flights and along the drum were simulated as binomial or trinomial random walk, and were validated by comparison with experiments and Monte Carlo simulations. Since the flight geometry makes the axial and cross-sectional material transport complicate, no predictive model has yet been developed.

4.4. Fluidized bed

Many recent publications on fluidized beds focused on circulating fluidized beds (CFB), which is equivalent to a batch reactor whereas the riser part can be considered as a continuous flow system. Chan et al. [48] focused on the velocity profile and the RTD of solid phase in the riser using PEPT technique. The existence of four operating solid hold-up zones was validated in this work at prevailing conditions of superficial air velocity and solid circulating flux. Mahmoudi et al. [13] investigated the back-mixing and the RTD of a gas phase in the riser of a CFB. Results in this study showed that dilute riser flow and dense riser upflow led to plug flow, whereas the core-annulus operation enhanced back-mixing. Stochastic models were developed for both four zones and core-annulus cases through the use of a Markov chain [42]. Rodríguez-Rojo et al. [12] introduced the application of fluidized bed on particle coating with supercritical carbon dioxide, where a well-mixed solid and supercritical fluid is the key point of good performance. The influences of gas and solid properties on the RTD were also analyzed. Andreux et al. [21] similarly studied the solid motion inside the riser of a CFB, focusing on the pressure drop due to the solid flux, which divided the vertical riser into the acceleration zone and the established zone. A decrease of solid axial dispersion was observed with the increase of solid flux rate, similar to the results in rotary drums. Other than the circulating solid flux and the superficial gas velocity, the influence of the riser exit geometry was also investigated [31,71]. The geometry showed a modest but consistent influence upon the particle RTD. When reaction was involved, a continuous combustor similar to CFB was studied and compared with similar batch tests [3]. From analysis of the gas leaving the air reactor, the RTD of fuel particles and the circulating solid flux in the air riser were determined in this reactor. The conversion rate and the limitation of this reaction were analyzed coupling with material transport.

5. Conclusion

The RTD method has been widely applied in industrial continuous flow systems. It offers a convenient tool for understanding material transport phenomena inside various unit operations, which is the first step for efficient operation design, troubleshooting, and system improvement.

Focusing on solid systems, this paper summarizes the industrial developments using the residence time theory since the last theoretical review of Nauman [7]. There remain many interesting aspects that benefit practical RTD applications and can be explored.

Efforts are required focusing on the connection between the RTD profile and the performance of different continuous unit operations. If the RTD of a continuous reaction system can be coupled with the reaction performance of an equivalent batch system, useful and efficient guidance can be provided on the design and the performance improvement of the corresponding continuous system, which directly benefits industrial use.

Most current literatures focused on the RTD profiles of various continuous systems, or modeling selection based on RTD fitting performance. Since no new modeling was actually developed in recent years, the value of these studies mainly relies on the RTD characteristics explored for the specific systems.

Efforts are recommended that not only apply existing fitting modeling of the RTD, but also provide novel predictive modeling. These efforts significantly benefit in the application of residence time theory because time and material cost on the RTD experiments can be saved. It is not very difficult to predict the MRT when a reasonable estimation of the material fill level or hold-up is achievable. However, the critical step in these efforts is the prediction of axial dispersion, due to the complicate influence of operations and material properties in different unit operations. An alternative of the predictive modeling is the application of CFD or DEM simulations, where validation and verification of the simulations with experimental results is always required. In this case, a valid correlation between practical parameters (operation, material properties, geometry etc.) and simulation parameters should be the focus in order to achieve reliable and applicable conclusions from the simulations.

The development of inline tracer detection method, especially for solid systems, is also a promising direction that requires more work.

Acknowledgement

This work is supported by the National Science Foundation Engineering Research Center on Structured Organic Particulate Systems, through grant NSF-ECC 0540855, and by grant NSF-0504497. The authors would also like to thank Douglas Hausner for editing this manuscript.

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